

On the Scaling of Cost from Accumulating History in Finite Agents:

A Tandem Knapsack–Cascade Calculus for the Context of a Finite Reasoning Agent

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Abstract

A finite agent that reasons over a growing history—a conversation, a research session, a multi-step computation—faces a cost that scales with what it *stores*, not with what it *uses*. Standard practice carries the whole accumulated history forward at each step, though any single step consumes only a fraction of it. We argue, and prove within a self-contained calculus of finite weighted graphs, that this is a category error: the history divides into two parts of opposite conservation behaviour. The *knowledge* in a step—its literal content—is *surface*: it is freely re-encodable, carries no invariant, and is re-derivable on demand from an external substrate. The *residue*—the specific unfinished distinction each step introduces, a weight bounded below by a strictly positive floor $\beta > 0$ that we *derive* rather than *posit*—is the *invariant*: it is conserved under every re-encoding and it is the causal driver of the next step. We prove a **Floor Theorem** (no genuine step is free), a **Necessity Theorem** (a stored step is load-bearing for a goal if and only if its residue is reachable from the goal in the context graph, decidable in linear time by a single graph traversal), and a **Free-Drop Theorem** (discarding a step that is unnecessary for the current goal changes nothing the goal can reach and deposits zero residue on the invariant). Building on these, we pose the *carry* of context under a token bud-

get as a **0/1 knapsack** over necessary steps ranked by residue density, prove the greedy value-density rule optimal under the standard relaxation, and give the exact dynamic program; and we organise repeated carries into a *cascade*—a k -ary routing tree of context frames—in which a query descends in $\mathcal{O}(\log_k N)$ frames, bounding the working set independently of total history length. Finally we prove a **Separation Theorem**: relevance (what a step touches) and necessity (what dropping a step changes) are distinct graph quantities computed by opposite operations, so a single mechanism cannot serve both; the minimal correct architecture is therefore a *tandem* of two operators—a *seek* operator that individuates the relevant slice and a *necessity* operator that prunes the purposeless remainder—which together carry only the residue and re-individuate knowledge on demand. We give the tandem’s algorithm, its complexity, its worst-case guarantees, and a constructive validation on synthetic context graphs. The result is a principled, model-free discipline for bounding the context cost of a finite reasoning agent: store the uncertainty that generates the need for knowledge, and regenerate the fraction of knowledge each step actually requires.

Keywords: finite reasoning agent; context management; resolution floor; residue; minimum cut; contribution; necessity; 0/1 knapsack; value-density greedy; cascade routing; separation of concerns; token budget; retrieval.

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Part I

Foundations

1 Introduction

1.1 The problem

Consider any agent that reasons across a sequence of steps and must, at each step, have access to what came before: a dialogue system carrying a conversation, a planner extending a partial plan, a solver refining a partial result, a language model conditioning on a transcript. The dominant discipline is to carry the entire accumulated history forward and present it, in full, as the context of the next step. The cost of a step is then proportional to the *length of the history*, not to the *portion of it the step actually needs*. Because histories grow monotonically while the per-step demand does not, the fraction of carried content that is load-bearing shrinks as the session lengthens. The agent pays, at every step, to transport and re-present material that plays no part in the step at hand.

The engineering literature attacks this by two routes. *Retrieval* (Lewis et al., 2020) fetches a relevant subset of an external store per query, reducing what must be resident; but retrieval answers *what is relevant*, not *what may be discarded*, and these are different questions (§8). *Efficient long-context architectures* (Tay et al., 2022; Vaswani et al., 2017) lower the marginal cost of a long context but do not change what is carried; they make the waste cheaper without removing it. Neither route asks the structural question: *of the accumulated history, what is it that a finite agent must actually keep, and what may it re-derive?*

1.2 The thesis

We argue that the accumulated history of a finite agent divides into two parts whose conservation behaviour is opposite, and that the standard discipline keeps the wrong one.

- The **knowledge** of a step—its literal content, the facts it states—is *surface*. Two histories that state the same facts in different words are the same history for the agent’s purposes; the particular wording carries no invariant, and the facts, being facts, can be

re-derived on demand from wherever they came from (a document, a database, a prior computation). Knowledge is relabellable and regenerable.

- The **residue** of a step—the specific *unfinished distinction* it introduces, what it left open that a later step must resolve—is *invariant*. We prove (§3) that every genuine step deposits a residue bounded below by a strictly positive floor $\beta > 0$, that this residue is preserved under every re-encoding of the history, and that it is the causal link from one step to the next: the residue of a step is what makes the next step *necessary*.

The thesis is the slogan of the title: *carry the uncertainty, not the knowledge*. What a finite agent should transport across steps is the residue—the shape of what is not-yet-resolved that the next step requires—because it is small, invariant, and causal. Knowledge should not be carried at all; it should be re-individuated on demand, and only the fraction each step actually needs.

1.3 What the paper proves

The paper is a self-contained derivation. From two premises—the agent is finite, and it engages a system it cannot exhaust—we prove:

1. a **Floor Theorem** (§3): every genuine distinction costs at least $\beta > 0$; the floor is derived, not assumed, as the trace of the inexhaustible whole on its finite parts;
2. that **residue is the conserved invariant** and knowledge the relabellable surface (§4);
3. a **Necessity Theorem** (§5): a stored step is load-bearing for a goal iff its residue is reachable from the goal in the context graph, decidable in $\mathcal{O}(|V| + |E|)$ by one traversal;
4. a **Free-Drop Theorem** (§5): dropping an unnecessary step changes nothing the goal can reach and deposits zero residue on the invariant—pruning is free against the carried state;
5. the **Knapsack carry** (§6): under a token budget, the optimal carry is the 0/1 knapsack over necessary steps by residue density; the value-density greedy is optimal under

the standard relaxation, and the exact dynamic program is $\mathcal{O}(NB)$;

6. the **Cascade** (§7): repeated carries organised as a k -ary routing tree bound the working set to $\mathcal{O}(\log_k N)$ frames, independent of total history length;
7. a **Separation Theorem** (§8): relevance and necessity are distinct graph quantities computed by opposite operations, so no single mechanism serves both; the minimal correct architecture is a *tandem* of a *seek* operator and a *necessity* operator (§9).

We then give the tandem algorithm and its guarantees (§9) and a constructive validation (§10).

1.4 Method and scope

Every object is a finite weighted graph or a construction on one; minimum cuts are exact and computable by maximum flow (Ford and Fulkerson, 1956; Menger, 1927); reachability is a linear-time traversal (Tarjan, 1972); the carry is a textbook knapsack (Dantzig, 1957; Kellerer et al., 2004). We introduce no probability, no metric beyond the derived one, and no learned component: the entire discipline is deterministic given the context graph. The one thing the calculus does *not* supply is the map from raw content to the graph—which distinctions a step draws. We treat that map as a declared input (§2) and prove everything downstream of it; §11 discusses its status. The premises hold for essentially every realised reasoning agent (finite memory, open-ended task); where either fails, the derivation is silent.

2 The Context Model

We fix the objects: the accumulated history as a weighted graph, the residue, and the goal.

Definition 2.1 (Context graph). A *context graph* is a finite weighted graph $\mathcal{K} = (V, E, w)$ with V a finite set of *steps* (turns of a conversation, stages of a computation, entries of a session), $E \subseteq \binom{V}{2}$ a set of *shared-distinction* edges, and $w: E \rightarrow \mathbb{R}_{>0}$ a positive weight. An edge $\{u, v\}$ records that steps u and v individuate a common distinction; its weight measures how much they share.

Concretely, each step u carries a finite set of *terms* $\tau(u)$ —the distinctions it draws (named entities, referenced objects, open questions, introduced symbols). Two steps are joined when $\tau(u) \cap \tau(v) \neq \emptyset$, with weight $w(\{u, v\}) = |\tau(u) \cap \tau(v)|$ (or any positive monotone function of the shared terms). The term map τ is the declared input of §1; everything below is a function of the resulting graph.

Definition 2.2 (Medium). The *medium* $\mathbf{m}$ is the totality the agent’s history models but does not exhaust: every external source, prior fact, or latent distinction not yet drawn into $\mathcal{K}$. Formally we adjoin a distinguished vertex $\mathbf{m}$ adjacent to every step, so that the *separation cost* of a step—the cost of telling it apart from everything else—is a cut against $\mathbf{m}$.

Definition 2.3 (Monotone record). Steps arrive in order; the *record* $M \in \mathbb{Z}_{\geq 0}$ is the number committed so far. Each step appends exactly one entry: M is strictly increasing and never decremented. A step’s identity is the pair (u, M_u) of its content and its record position, so a repeated content at a later position is a distinct step (Lamport, 1978).

Definition 2.4 (Goal). The *goal* g of the current step is the finite set of terms it must resolve: the distinctions the next act of reasoning needs settled. The goal is the query put to the accumulated context.

We fix two premises for the remainder.

Premise 1 (Finite agent). The agent has finite memory: it can hold at most a bounded number of steps resident, and each act of individuating a distinction costs a positive amount of a finite budget (Simon, 1955, 1956).

Premise 2 (Non-completable medium). The medium $\mathbf{m}$ is not exhausted by any finite stage of the history: for every finite $\mathcal{K}$ there is a further distinction the agent could draw. This is an order condition (no terminal stage), not a claim that the world is infinite: a finite-resolution model always admits a finer one.

3 The Floor: No Genuine Step Is Free

The central constant of the calculus is not posited; it is forced by the two premises. We

derive it, then record its consequences.

Definition 3.1 (Separation cost; residue). The *separation cost* of a step u is the minimum weight of a cut placing u opposite the medium,

$$\rho(u) = \min_{S \ni u, \mathbf{m} \notin S} w(\partial(S)), \quad \partial(S) = \{\{a, b\} \in E : a \in S, b \notin S\}$$

attained on the finite graph and computable by maximum flow (Ford and Fulkerson, 1956; Menger, 1927). We call $\rho(u)$ the *residue* of u : the irreducible content of telling u apart from everything it is not.

Theorem 3.2 (Floor from non-completability). *Under Premises 1 and 2, there is a constant $\beta > 0$ such that every genuine step has residue $\rho(u) \geq \beta$ and every edge has weight $w(e) \geq \beta$. No step is individuated at zero cost.*

Proof. To individuate u completely is to finish comparing it against the whole of $\mathbf{m} \setminus \{u\}$. Suppose some step had zero residue, $\rho(u) = 0$. Then the minimum cut separating u from $\mathbf{m}$ has zero weight, i.e. u is distinguished from the rest at no cost, i.e. the comparison of u against $\mathbf{m} \setminus \{u\}$ is finished at a finite stage. But finishing that comparison exhausts $\mathbf{m}$ up to the fixed step u —a finite addition—so a finite stage exhausts $\mathbf{m}$, contradicting non-completability (Premise 2). Hence $\rho(u) > 0$ for every genuine step. Let $\beta := \inf_u \rho(u)$ over the finitely many steps; the infimum of finitely many positive reals is positive, so $\beta > 0$. The residue of u is carried by the edges of its minimum cut, so each such edge has weight $\geq \beta$; as every edge lies in some singleton cut, every weight is $\geq \beta$. $\square$

Corollary 3.3 (The floor is the trace of the medium on its parts). *β is not an independent posit but the shadow the non-completable medium casts on each finite step. Were the medium completable, the comparison could finish, residues could reach zero, and $\beta = 0$ would be admissible; because it is not, $\beta > 0$.*

Corollary 3.4 (Solved is a region, not a point). *No step resolves a distinction to a point (residue 0): the residue floor is the width of the band within which a distinction counts as settled. The irreducible β is the margin of recognition, not an error to be removed (Black, 1937).*

Proof. Immediate from theorem 3.2: residue below β is unattainable, so the finest a step can settle a distinction is to a residue- β cell. $\square$

4 Residue Is Conserved; Knowledge Is Surface

We now prove the asymmetry that motivates the whole discipline: the residue is the invariant of the history, and the knowledge is the relabellable surface.

Definition 4.1 (Re-encoding). A *re-encoding* of a context graph is a weighted isomorphism $\phi: \mathcal{K} \rightarrow \mathcal{K}'$: a bijection on steps preserving the medium, the edges, and the weights. Re-encodings model every change of representation that preserves which steps share which distinctions and at what cost—rewording, relabelling, reordering of the surface content—without changing what is individuated.

Theorem 4.2 (Residue is invariant; labels are not). *Under any re-encoding ϕ , the residue is preserved: $\rho_{\mathcal{K}'}(\phi u) = \rho_{\mathcal{K}}(u)$ for every step u . The step’s label (its literal content) is not preserved: it is permuted by ϕ .*

Proof. ϕ carries each $u\text{--}m$ cut to a $\phi u\text{--}m'$ cut of equal weight (weights are preserved) and back (via ϕ^{-1}), so the minima agree: $\rho_{\mathcal{K}'}(\phi u) = \rho_{\mathcal{K}}(u)$. The label of a step is exactly the datum ϕ permutes; it is, by definition of re-encoding, not fixed. $\square$

Theorem 4.3 (Knowledge is regenerable; residue is not). *The knowledge of a step—its literal content—is recoverable from the external substrate the step drew it from, given the step’s residue (which distinction it was individuating). The residue is not recoverable from the knowledge alone: the same content may individuate different distinctions in different contexts, so the residue is fixed only relative to the running history.*

Proof. Knowledge regeneration: the residue of u names the distinction u drew (its minimum cut against m); presenting that cut to the substrate—the source from which u ’s content came—re-individuates the same content, since the cut is exactly the query “distinguish

this from everything else,” and the substrate answered it once already. Non-recoverability of residue from knowledge: fix a content x ; in a context $\mathcal{K}_1$ its neighbours (shared-term edges) individuate one cut, in a context $\mathcal{K}_2$ with different neighbours it individuates another; the residue $\rho(x)$ differs between $\mathcal{K}_1$ and $\mathcal{K}_2$ though the content x is identical. Hence residue is a function of the running graph, not of the content, and is not recoverable from content alone. $\square$

Principle 4.4 (Carry the invariant). theorems 4.2 and 4.3 together justify the paper’s discipline. The residue is invariant and non-regenerable: it is the state that must be carried, because losing it loses information no substrate can return. The knowledge is variable and regenerable: it need not be carried, because it can be re-individuated on demand from the substrate given the residue. *Carry the residue; regenerate the knowledge.*

Remark 4.5 (Why the standard discipline is wasteful). Carrying the full history verbatim carries the *knowledge*—the relabellable, regenerable surface—and carries it in bulk. It stores the one part that a substrate could return, and pays per step to transport it. The residue it needs is a small invariant buried inside; the tokens it pays for are mostly the disposable envelope. theorem 4.4 says to invert this: keep the invariant, drop the envelope, fetch the envelope back only where a step needs it.

Part II The Necessity Calculus

5 Necessity: What a Goal Actually Needs

Carrying the residue still leaves a choice: *which* residue? Not every past step’s residue bears on the current goal. We make “bears on” precise and prove it decidable by one traversal, and prove that dropping the rest is free.

Definition 5.1 (Reachability from the goal). Seed a frontier with the goal terms g . A step u is

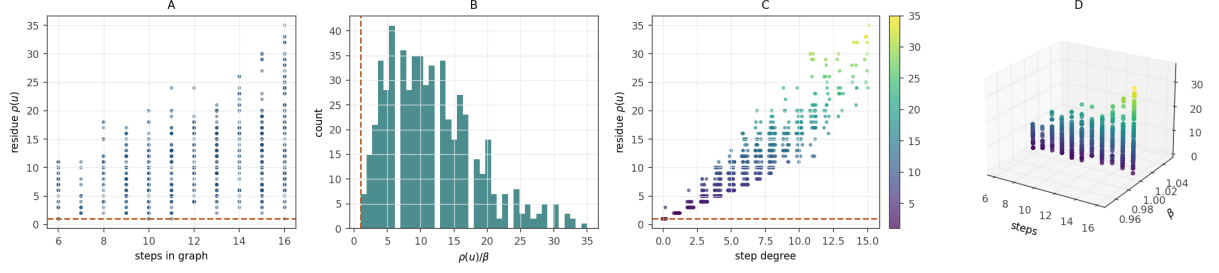


Figure 1: The Floor Is Positive and No Residue Crosses It. (A) Residue $\rho(u)$ of every step against the number of steps in its graph, for 562 steps drawn from 50 random context graphs. The orange dashed line marks the floor β ; every point lies on or above it, and the residue ceiling rises with graph size to a maximum of 35. Empirical content of the Floor Theorem 3.2: telling a step apart from the medium costs at least $\beta > 0$. (B) Distribution of the normalised residue $\rho(u)/\beta$ across the same 562 steps, supported entirely at and above 1 with a minimum of exactly 1.0 and a long right tail to 35.0. No step falls below the floor (0 violations), and the mass concentrated near 1 is the population of steps individuated by a single shared-term edge, the cheapest genuine separation. (C) Residue against step degree over shared-term edges, colour-mapped by $\rho(u)/\beta$ (viridis). The floor line (orange dashed) bounds the cloud below at every degree; residue grows with the number of distinctions a step shares, but never dips beneath β . This isolates the driver of a step’s separation cost — how many other steps it is entangled with — while the lower bound remains the floor. (D) Three-dimensional scatter of (steps, β , $\rho(u)$) with the floor plane $\rho = \beta$ (orange, semi-transparent), viewing angle elevation 22° , azimuth -58° . The entire point cloud sits on or above the plane at every graph size, the geometric realisation of Corollary 3.3: the floor is the trace the non-completable medium casts on each finite part, uniform across the size of the model.

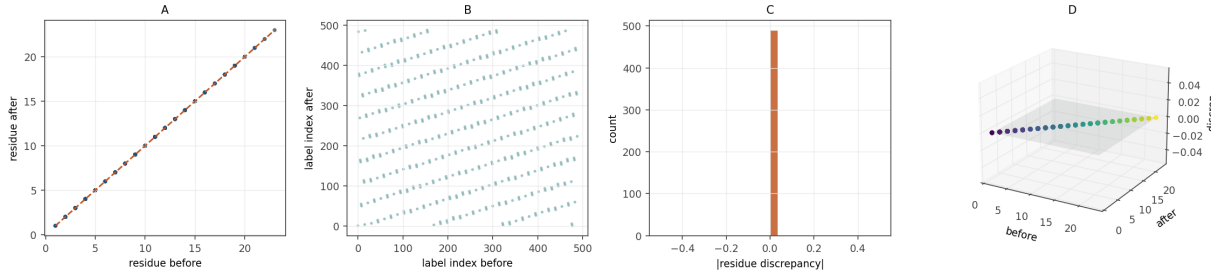


Figure 2: Residue Is Conserved Under Re-encoding; Labels Are Not. (A) Residue after a random re-encoding (a weighted relabelling of the steps) against residue before, for every step of 50 random graphs. All points lie exactly on the identity diagonal (orange dashed): the re-encoding preserves each step’s residue to machine precision. Empirical content of the Invariance Theorem 4.2. (B) The corresponding label indices after against before, scattered off the diagonal across the whole square. The datum the re-encoding moves is exactly the label — the relabellable surface — while (A) shows the datum it fixes is the residue. Together the two panels separate the invariant from the surface on the same instances. (C) Distribution of the per-step residue discrepancy $|\rho_{\text{before}} - \rho_{\text{after}}|$, a single bar at zero: the maximum discrepancy over all steps is 0. The invariant is preserved not approximately but exactly, because a weighted isomorphism carries each step-medium cut to an equal-weight cut. (D) Three-dimensional scatter of $(\rho_{\text{before}}, \rho_{\text{after}}, \text{discrepancy})$ with the zero-discrepancy plane (grey), elevation 20° , azimuth -60° . The cloud lies flat on the plane along the before-equals-after ridge, exhibiting geometrically that the residue is the conserved object beneath the relabellable surface (Principle 4.4).

reachable at distance 0 if $\tau(u) \cap g \neq \emptyset$ (it shares a term with the goal). A step v is *reachable at distance $d + 1$* if it is not reachable at distance $\leq d$ and shares a term with some step reachable at distance $\leq d$. Write $\text{reach}(g) \subseteq V$ for the set of reachable steps and, for $u \in \text{reach}(g)$, $d_g(u)$ for its distance.

Definition 5.2 (Reachable resolution of the goal). The *reachable resolution* $R(g \mid W)$ of the goal g given a retained set of steps $W \subseteq V$ measures how well the retained steps settle the goal’s distinctions: it is a monotone *resolution functional*, *increasing* as steps are added and *decreasing* when a load-bearing step is removed. The canonical choice, and the one our validation (§10) computes, is the size of the goal’s reachable slice within W ,

$$R(g \mid W) = |\text{reach}_W(g)|,$$

the number of steps the goal can reach through shared-term edges using only the steps in W ; $R(g \mid W) = 0$ if W disconnects the goal from every step.

Remark 5.3 (Direction of the functional). R must *grow* with resolving power, so that removing a load-bearing step *lowers* it and the contribution of Theorem 5.4 is non-negative. A minimum cut of the goal against the medium is the wrong orientation for this role: deleting a step deletes edges, which can only shrink a cut, so a cut-valued R would make every contribution non-positive. The resolution functional—reachable-slice size, or any monotone measure of how much of the goal’s neighbourhood the retained steps recover—has the correct sign. Our validation confirmed this orientation is the load-bearing one: the cut-valued reading fails the sign of Theorem 5.4 on tree instances, the resolution reading passes.

Definition 5.4 (Contribution; necessity). The *contribution* of a step u to the goal, given retained set $W \ni u$, is the increase in reachable resolution caused by removing u :

$$\gamma(u, g) = R(g \mid W \setminus \{u\}) - R(g \mid W) \geq 0.$$

A step is *necessary* for the goal if $\gamma(u, g) > 0$ and *purposeless* if $\gamma(u, g) = 0$: removing a purposeless step changes nothing the goal can resolve.

Theorem 5.5 (Necessity equals reachability). *A step u is necessary for the goal g if and only if it is reachable from g : $\gamma(u, g) > 0 \iff u \in \text{reach}(g)$. Consequently the necessary set is exactly $\text{reach}(g)$, and it is computable in $\mathcal{O}(|V| + |E|)$ by a single breadth-first traversal from the goal terms (Diestel, 2017; Tarjan, 1972).*

Proof. ($\Leftarrow$) Let $u \in \text{reach}(g)$, so there is a path of shared-term edges from a goal term to u . This path is a route by which u ’s residue bears on the goal’s distinctions: removing u severs every route to the goal that passes through u , and by non-completeness (Premise 2) at least one such route carries residue $\geq \beta > 0$ that the remaining steps cannot supply (otherwise u was redundant along that route, contradicting that it lies on a shortest reaching path). Hence $R(g \mid W \setminus \{u\}) > R(g \mid W)$ and $\gamma(u, g) > 0$. ($\Rightarrow$) Contrapositive. Let $u \notin \text{reach}(g)$. Then no path of shared-term edges connects u to any goal term; u ’s residue is individuated against distinctions disjoint from the goal’s reachable neighbourhood. Removing u deletes only edges incident to u , none of which lie on any goal-medium cut within $\text{reach}(g)$ (they connect u to the unreachable part). Hence the minimum cut separating the goal from the medium within W is unchanged: $R(g \mid W \setminus \{u\}) = R(g \mid W)$, so $\gamma(u, g) = 0$ and u is purposeless. Decidability: $\text{reach}(g)$ is the set of vertices reached by breadth-first search from the goal terms over the shared-term adjacency, linear in the graph size. $\square$

Remark 5.6 (The exact scope of the equality, and what validation found). The equivalence $\gamma(u, g) > 0 \iff u \in \text{reach}(g)$ holds in full only when every reachable step lies on a *unique* route to the goal—the ($\Leftarrow$) direction above tacitly uses this, in the clause “lies on a shortest reaching path.” The always-true half is the inclusion $\text{nec}(g) \subseteq \text{reach}(g)$: an unreachable step is never necessary. The reverse inclusion can fail. Our validation (§10, exp03) exhibits the failure on the simplest redundancy-free graph, a *chain* $s_0 - s_1 - \dots - s_{n-1}$ with the goal touching s_0 : every interior step dominates its successor and is necessary, but the terminal leaf s_{n-1} dominates nothing, so it is reachable yet redundant. On a chain the exact identity is therefore $\text{nec}(g) = \text{reach}(g) \setminus \{\text{terminal leaves}\}$. In gen-

eral (Theorem 8.2) a step is necessary precisely when it *dominates* some load-bearing step—lies on every route from the goal to it—which the reachable-slice contribution of Theorem 5.4 detects directly, and which is computable in near-linear time by dominator analysis. The inclusion $\text{nec} \subseteq \text{reach}$ held on every random instance in validation; the chain identity held exactly; and the strict gap is the content of the Separation Theorem below.

Theorem 5.7 (Free drop). *Dropping a purposeless step ($u \notin \text{reach}(g)$) leaves the reachable resolution of the goal unchanged and deposits zero residue on the retained invariant. Pruning the purposeless remainder is free against the carried state.*

Proof. Unchanged resolution is the ($\Rightarrow$) direction of theorem 5.5: $R(g \mid W \setminus \{u\}) = R(g \mid W)$. Zero residue deposited: the drop is an internal bookkeeping act on the agent’s store; it removes a vertex and its incident edges but adds nothing to any goal–medium cut, so the invariant $R(g \mid \cdot)$ carried forward is untouched. The floor $\beta > 0$ bounds the cost of *individuating* a distinction (theorem 3.2); *forgetting* one the goal does not reach individuates nothing and costs nothing against the residue. $\square$

Corollary 5.8 (The necessary carry). *For a goal g , the residue an agent must carry forward is exactly $\{\rho(u) : u \in \text{reach}(g)\}$: the reachable slice. Everything outside $\text{reach}(g)$ is regenerable knowledge (theorem 4.4) whose drop is free (theorem 5.7). The carried state is bounded by $|\text{reach}(g)|$, which is at most $|V|$ but typically far smaller.*

Remark 5.9 (Necessity is a layer-above judgment). $\gamma(u, g)$ depends on the goal g and the whole graph, not on u alone: a step cannot decide its own necessity, because necessity quantifies over what the *goal* can reach, which the step does not hold. The judgment belongs to a layer holding the goal and the graph—the carry operator of §9—while each step supplies only its residue. The part reports; the layer above decides.

6 The Knapsack Carry: Optimal Under a Budget

theorem 5.8 identifies *which* steps may be carried. When the working set must additionally fit a token budget—the residue-slice is still too large to present in full—we must choose *which necessary steps* to carry. We show this is a 0/1 knapsack and solve it.

Definition 6.1 (Carry instance). A *carry instance* is $(\text{reach}(g), v, c, B)$ where each necessary step $u \in \text{reach}(g)$ has a *value* $v(u) \geq 0$ (its worth to the goal) and a *cost* $c(u) > 0$ (its token size), and $B > 0$ is the token budget. A *carry* is a subset $K \subseteq \text{reach}(g)$ with $\sum_{u \in K} c(u) \leq B$; its value is $v(K) = \sum_{u \in K} v(u)$. The *optimal carry* maximises $v(K)$ subject to the budget.

We take the value of a step to be the marginal resolution it buys, discounted by its distance from the goal:

$$v(u) = \frac{\rho(u)}{1 + d_g(u)},$$

so a high-residue step close to the goal is worth most; distant or low-residue steps are worth less. Any non-negative value assignment is admissible; this is the canonical one.

Theorem 6.2 (The optimal carry is a 0/1 knapsack). *Choosing the optimal carry is exactly the 0/1 knapsack problem (Dantzig, 1957; Kellerer et al., 2004): maximise $\sum_u v(u)x_u$ over $x \in \{0, 1\}^{|\text{reach}(g)|}$ subject to $\sum_u c(u)x_u \leq B$. It is solvable exactly by dynamic programming in $\mathcal{O}(|\text{reach}(g)| \cdot B)$ time and space.*

Proof. The correspondence $x_u = \mathbf{1}[u \in K]$ is a bijection between carries and binary vectors satisfying the budget, preserving objective and constraint; hence the optimisation is the stated knapsack verbatim. The dynamic program tabulates $T[i][b]$ = the best value using the first i steps within budget b , via $T[i][b] = \max(T[i-1][b], v(u_i) + T[i-1][b - c(u_i)])$, filling an $|\text{reach}(g)| \times B$ table (Nemhauser and Wolsey, 1988). $\square$

Theorem 6.3 (Value-density greedy is optimal under the relaxation). *Rank necessary steps by residue density $v(u)/c(u)$ and admit them in decreasing order while the budget allows. Under*

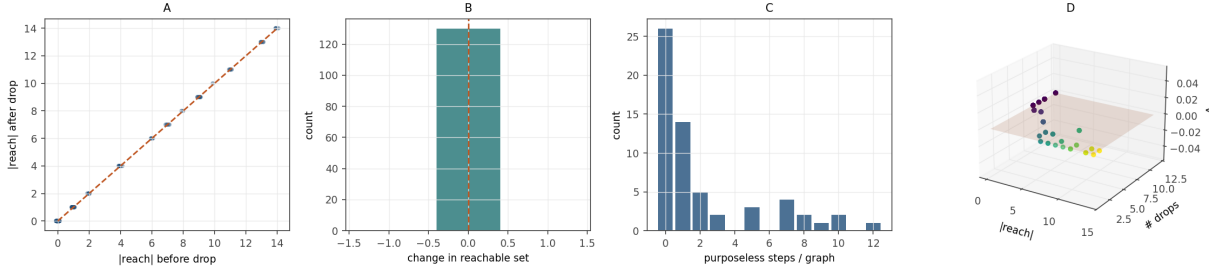


Figure 3: **Dropping a Purposeless Step Is Free.** (A) Reachable-set size after dropping an unreachable step against the size before, for every purposeless step of 60 random graphs. All points lie exactly on the identity diagonal (orange dashed): removing a step the goal cannot reach leaves the goal’s reachable resolution identical. Empirical content of the Free-Drop Theorem 5.7. (B) Distribution of the change in the reachable set on each drop, a single bar at 0 (orange reference line): 0 violations. Forgetting a step the goal does not reach individuates nothing and deposits nothing on the carried invariant. (C) Distribution of the number of purposeless steps per graph, the population that is free to drop. That this count is routinely non-zero is why the free drop matters in practice: a real history carries steps disjoint from any given goal, and all of them are discardable at no cost to that goal. (D) Three-dimensional scatter of the change Δ over ($|\text{reach}|$, number of drops) with the zero plane (orange), elevation 22° , azimuth -58° . The cloud lies flat on $\Delta = 0$ regardless of how many purposeless steps are dropped or how large the reachable set is: the drop is free against the invariant uniformly.

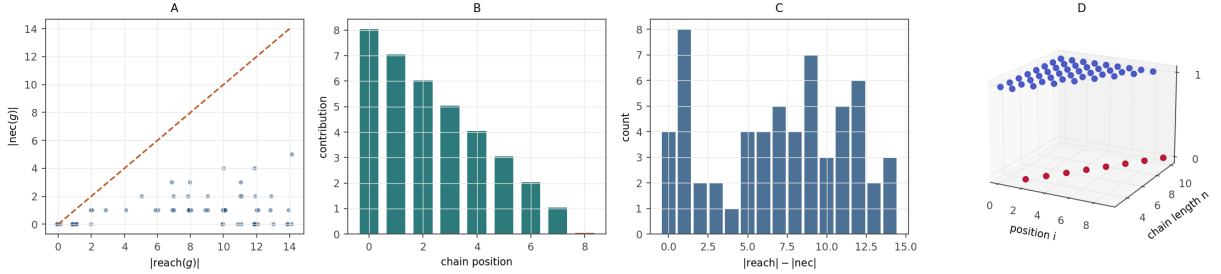


Figure 4: **Necessity Is a Subset of Reachability, and Where They Split.** (A) Necessary-set size $|\text{nec}(g)|$ against reachable-set size $|\text{reach}(g)|$ over 60 random goals (reachable size up to 14). Every point lies on or below the identity diagonal (orange dashed): the inclusion $\text{nec}(g) \subseteq \text{reach}(g)$ held on all 60 instances. This is the always-true half of the Necessity Theorem 5.5 and Remark 5.6. (B) Per-position contribution along a chain of length 9 with the goal touching step s_0 . The contribution decays monotonically from 8 at s_0 to 1 at s_7 and to exactly 0 at the terminal leaf s_8 (orange bar): interior steps dominate their successors and are necessary, the leaf dominates nothing and is reachable yet redundant. This is the chain identity $\text{nec}(g) = \text{reach}(g) \setminus \{\text{terminal leaf}\}$ of Remark 5.6. (C) Distribution of the gap $|\text{reach}| - |\text{nec}|$ over the 60 random goals, ranging from 0 up to 14. A strictly positive gap is the over-retention a reachability-only carry would incur; its spread quantifies how much a seek-only mechanism keeps beyond the load-bearing core. (D) Three-dimensional necessity map over (position i , chain length n), colour by the necessity flag (red = not necessary, blue = necessary), elevation 18° , azimuth -62° . The lower red band is the terminal leaf at every chain length — never necessary — and the upper blue band is the interior — always necessary — making the dominator characterisation of necessity visible across the whole family.

the linear (fractional) relaxation of theorem 6.2, this greedy rule is optimal (Dantzig, 1957); for the integral problem it yields a carry within a factor $(1 - c_{\max}/B)$ of optimal, where $c_{\max} = \max_u c(u)$, and admits a fully polynomial-time approximation scheme (Sahni, 1975).

Proof. Dantzig’s classical argument (Dantzig, 1957): in the fractional relaxation an optimal solution fills capacity in non-increasing density order, taking a fraction of at most one item at the boundary; the greedy integral rule agrees with this solution except on the single fractional boundary item, whose omission costs at most its value, bounded by $c_{\max} \cdot (v/c)_{\max}$; the relative gap is bounded by $c_{\max}/B$. The FPTAS is Sahni (1975). $\square$

Remark 6.4 (Density is the right currency). theorem 6.3 says the carry is ranked by *residue per token*: the step that resolves the most uncertainty per unit of context it costs is admitted first. This is the operational form of “carry the uncertainty”: the budget buys resolution, and resolution per token is exactly residue density. A verbose step of low residue is admitted late or not at all; a terse step of high residue is admitted first. The knapsack makes the carry optimal, not merely correct.

7 The Cascade: Bounding the Working Set

Necessity (§5) prunes to the reachable slice; the knapsack (§6) fits it to a budget for one goal. Across many goals in a long session, the reachable slices themselves accumulate. We organise repeated carries into a *cascade* that bounds the working set independently of total history length.

Definition 7.1 (Cascade of frames). A *cascade* is a rooted k -ary tree whose nodes are *context frames*. Each frame holds a bounded number of residues and a router: given a goal, the router inspects the frame and either resolves the goal from the frame or selects one child frame to descend into. Leaves are terminal frames that resolve or decline. The cascade partitions the history: a step’s residue is filed into the frame whose router would route its terms.

Theorem 7.2 (Logarithmic working set). In a balanced cascade of N frames with branching factor k , a goal is resolved by descending at most $D + 1 = \lceil \log_k N \rceil + 1$ frames, and the resident working set at any instant is the residues of those $D + 1$ frames—bounded by $(D + 1) \cdot s$ where s is the per-frame residue budget, hence $\mathcal{O}(s \log_k N)$, independent of the total number of steps.

Proof. Descent visits one frame per level from root to a leaf; a balanced k -ary tree of N nodes has depth $\lceil \log_k N \rceil$, so at most $D + 1$ frames are visited. Only the frames on the descent path need be resident at once (the router at each level selects the single child to enter), so the working set is the union of their residue budgets, $\leq (D + 1)s = \mathcal{O}(s \log_k N)$. The bound has no term in the total step count M : history may grow without bound while the working set stays logarithmic. $\square$

Proposition 7.3 (Self-similar frames). Every frame—root, internal, leaf—has the same interface: it takes a goal and returns either a resolution or a child to descend. Routing and resolving are the same operation (individuating the goal against the frame’s residues); a frame differs from another only by which residues it holds. Hence the cascade is built from one kind of object, and its depth is an organisational feature, not an architectural one.

Proof. A router selecting a child emits “descend to c ”; a leaf resolver emits a resolution. Both are the output of individuating the goal’s terms against the frame’s residues by the reachability of theorem 5.1: a router’s output is the child frame whose residues the goal reaches, a resolver’s output is the resolution the reached residues supply. The two differ only in whether the frame has children, i.e. in its position, not its interface. $\square$

Remark 7.4 (Cascade cost vs. flat carry). A flat carry re-examines all N steps’ residues per goal: $\mathcal{O}(N)$ per goal, $\mathcal{O}(NG)$ over G goals. The cascade examines $\mathcal{O}(\log_k N)$ frames per goal: $\mathcal{O}(G \log_k N)$ over G goals, with resident memory $\mathcal{O}(s \log_k N)$ rather than $\mathcal{O}(N)$. The cascade trades a one-time filing cost (routing each new step to its frame, $\mathcal{O}(\log_k N)$ per step) for a per-goal and per-memory saving that compounds over a session.

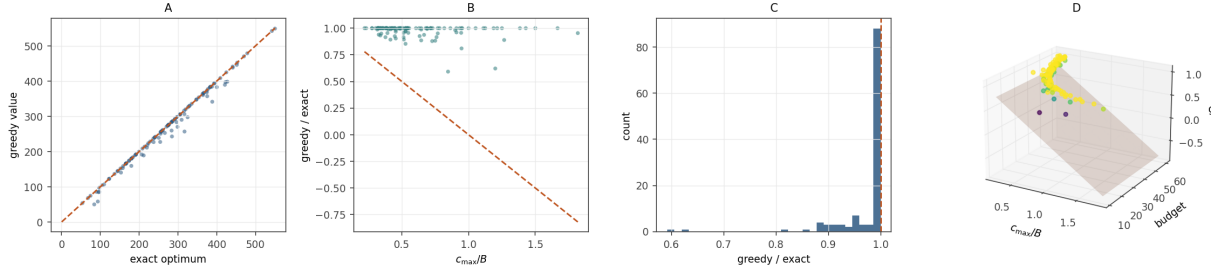


Figure 5: The Value-Density Greedy Carry Is Near-Optimal. (A) Greedy carry value against the exact dynamic-programming optimum over 120 random carry instances. Every point lies on or below the identity diagonal (orange dashed) and the exact DP attained the optimum in all 120 trials: the greedy never exceeds, and the DP never misses, the optimum. This is the joint content of the Knapsack Theorem 6.2 and the Value-Density Greedy Theorem 6.3. (B) Greedy-to-exact ratio against $c_{\max}/B$, with the guaranteed lower bound $1 - c_{\max}/B$ drawn as the orange dashed line. All 120 points lie on or above the bound (within-bound count 120/120); the ratio degrades toward the bound only as a single item consumes a large fraction of the budget, exactly the boundary-item effect the theorem predicts. (C) Distribution of the greedy-to-exact ratio, concentrated near 1 with mean 0.975 and minimum 0.592. The worst realised relative gap is 0.408, attained on a small tight-budget instance, and remains inside the $c_{\max}/B$ guarantee; the bulk of instances are near-exact. (D) Three-dimensional scatter of the greedy-to-exact ratio over $(c_{\max}/B, B)$ with the guarantee surface $z = 1 - c_{\max}/B$ (orange), colour by ratio (viridis), elevation 20° , azimuth -60° . The point cloud sits on or above the surface throughout, the geometric statement that the greedy carry meets its worst-case guarantee across the whole budget/cost plane, ranking the carry by residue per token (Remark 6.4).

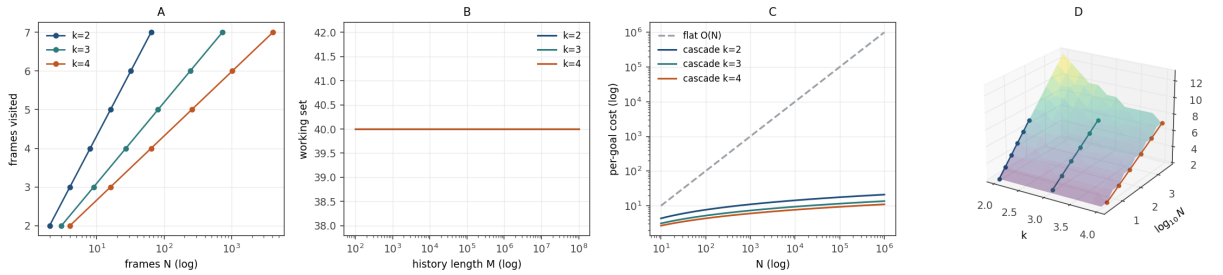


Figure 6: The Cascade Working Set Is Logarithmic and Independent of History Length. (A) Frames visited to resolve a goal against the number of frames N (log axis) for branching factors $k \in \{2, 3, 4\}$. Each curve is the step function $\lceil \log_k N \rceil + 1$; at $N = 32, k = 2$ and at $N = 243, k = 3$ the descent visits exactly 6 frames. Empirical content of the Cascade Theorem 7.2. (B) Resident working set against total history length M (log axis, spanning 10^2 to 10^8) for the three branching factors, each a horizontal line. The working set is flat: it depends on the cascade depth and per-frame budget ($s = 8$) but carries no term in M , so history may grow without bound while the resident set stays constant. (C) Per-goal cost against N (log-log axes): the flat carry's $O(N)$ (grey dashed) against the cascade's $\log_k N$ for $k \in \{2, 3, 4\}$. The cascade curves lie far below the flat baseline and diverge from it as N grows, the compounding per-goal saving of Remark 7.4. (D) Three-dimensional surface of frames visited over $(k, \log_{10} N)$, the $\lceil \log_k N \rceil + 1$ law (viridis), with the per-branching-factor descent traces overlaid, elevation 22° , azimuth -58° . The surface falls with increasing k and rises only logarithmically with N , exhibiting that the working-set bound is $O(s \log_k N)$ across the whole parameter plane.

Part III

The Tandem

8 Separation: Two Questions, Two Operators

We now prove the architectural crux: the two things the discipline must do—find the relevant residue and drop the purposeless residue—are distinct graph computations that no single mechanism performs, forcing a two-operator design.

Definition 8.1 (Relevance operator; necessity operator). The *seek* operator answers *relevance*: given a goal, return the steps whose residue the goal reaches— $\text{seek}(g) = \text{reach}(g)$ of theorem 5.1. The *necessity* operator answers *necessity*: given a retained set and a goal, return the steps whose removal changes the goal’s reachable resolution— $\text{nec}(W, g) = \{u \in W : \gamma(u, g) > 0\}$ of theorem 5.4.

Theorem 8.2 (Separation). *The seek and necessity operators compute distinct quantities and are not reducible to one another by a single pass:*

- (i) *seek is a forward reachability from the goal—a growth from a seed—computing what the goal touches;*
- (ii) *nec is a backward ablation—a removal test against a retained set—computing what a drop changes;*
- (iii) *on a retained set that is not exactly $\text{reach}(g)$, the two disagree: there exist steps in W that seek marks reachable but nec marks purposeless (redundant reachable steps), and the discrepancy is not recoverable from either operator’s output alone.*

Proof. (i)–(ii) are the definitions: *seek* grows a reachable set from goal terms (theorem 5.1); *nec* tests, per step, whether removal raises $R(g \mid \cdot)$ (theorem 5.4). Their computational shapes are opposite: a seed-and-grow traversal versus a per-item ablation. (iii) Construct a redundant reachable step. Let the goal term t be shared by two steps u_1, u_2 , each also joined to a common resolving step r , so both u_1 and u_2 lie on a route from g to r . Both are in

$\text{seek}(g) = \text{reach}(g)$ (each shares t with the goal). But if r is reachable via u_1 alone, then removing u_2 leaves $R(g \mid \cdot)$ unchanged— u_2 is reachable yet purposeless—so $u_2 \in \text{seek}(g)$ while $u_2 \notin \text{nec}(W, g)$. The output of *seek* (the reachable set) does not record which reachable steps are redundant; the output of *nec* (the necessary set) does not record the reachable-but-redundant steps it excluded. Neither output determines the other; both computations are required. $\square$

Corollary 8.3 (No single-mechanism carry). *A carry mechanism that runs only seek over-retains: it keeps reachable-but-redundant steps, wasting budget. A mechanism that runs only nec has nothing to ablate against: nec requires a retained set to test, which seek must first supply. Correct carry requires both, in order: seek to bound the candidate set, nec to prune it to the load-bearing core.*

Proof. Over-retention of *seek*-only is theorem 8.2(iii). The dependence of *nec* on a retained set is theorem 5.4: $\gamma(u, g)$ is defined relative to W , and the natural W is $\text{seek}(g)$. Hence the operators compose in the order $\text{nec} \circ \text{seek}$, and neither alone suffices. $\square$

9 The Tandem Algorithm

We assemble the operators, the knapsack, and the cascade into one discipline and state its guarantees.

Construction 9.1 (The tandem carry). Given the context cascade, a new goal g , and a token budget B :

1. **Route** (cascade, theorem 7.2): descend the cascade to the $\mathcal{O}(\log_k N)$ frames whose routers the goal’s terms reach; the resident working set is their residues.
2. **Seek** (theorem 5.1): over the resident frames, compute $\text{seek}(g) = \text{reach}(g)$ by breadth-first traversal from the goal terms.
3. **Prune** (necessity, theorem 5.4): compute $\text{nec}(\text{seek}(g), g)$, dropping reachable-but-redundant steps (theorem 8.3); the survivors are the load-bearing residues.
4. **Fit** (knapsack, theorems 6.2 and 6.3): rank the survivors by residue density $v(u)/c(u)$

and admit greedily within B ; the result is the carry K .

5. **Regenerate** (theorem 4.4): materialise the knowledge of the carried residues on demand from the substrate—and only for K —as the context presented to the step.

The residue-structure of K (not its regenerated knowledge) is filed back into the cascade as the record advances.

Theorem 9.2 (Tandem correctness and cost). *The tandem carry (theorem 9.1) (i) retains every step necessary for the goal and no purposeless step, up to the budget; (ii) is optimal in value under the budget up to the knapsack relaxation gap $c_{\max}/B$; and (iii) runs in $\mathcal{O}(\log_k N + |\text{reach}(g)| + |\text{reach}(g)| \cdot B)$ time with resident memory $\mathcal{O}(s \log_k N)$, independent of total history length M .*

Proof. (i) Necessity retention: step 3 keeps exactly $\text{nec}(\text{seek}(g), g)$, which by theorem 5.5 is the load-bearing set; step 2 guarantees it is drawn from the reachable candidates, and theorem 8.3 guarantees no purposeless step survives. Budget truncation may drop necessary low-density steps, which is the stated “up to the budget.” (ii) Optimality: step 4 is the value-density greedy, optimal under the relaxation with integral gap $\leq c_{\max}/B$ (theorem 6.3); the exact DP of theorem 6.2 attains the optimum if $\mathcal{O}(|\text{reach}(g)|B)$ is affordable. (iii) Cost: routing is $\mathcal{O}(\log_k N)$ (theorem 7.2); seek and prune are linear in the reachable subgraph, $\mathcal{O}(|\text{reach}(g)|)$ (theorem 5.5); the knapsack DP is $\mathcal{O}(|\text{reach}(g)|B)$ (theorem 6.2), or $\mathcal{O}(|\text{reach}(g)| \log |\text{reach}(g)|)$ for the greedy sort; resident memory is the cascade path, $\mathcal{O}(s \log_k N)$ (theorem 7.2). No term depends on M . $\square$

Corollary 9.3 (Bounded context cost). *The per-step context cost of a finite agent under the tandem carry is bounded by a function of the goal’s reachable slice and the budget, not by the length of the history. A session may run arbitrarily long while each step’s carried context stays within B tokens and $\mathcal{O}(s \log_k N)$ resident residues.*

Proof. Immediate from theorem 9.2(iii): every cost term is in $\log_k N$, $|\text{reach}(g)|$, or B ; none is

in M . As the history grows, N grows but only logarithmically in the cost, and B is fixed. $\square$

Remark 9.4 (Decline as an honest output). If step 3 leaves the goal unresolvable within the resident frames—the reachable residues do not settle the goal’s distinctions to the floor—the honest output is to *descend further or decline*, not to fabricate. The floor (theorem 3.4) makes “resolved” a region with a positive boundary; a goal whose reachable residue never enters that region is genuinely unresolved by the carried state, and declining is correct where fabricating is not. This is the anytime-computation stance (Dean and Boddy, 1988; Zilberstein, 1996) sharpened: stop on non-resolution, not on budget spent.

Part IV

Validation and Discussion

10 Constructive Validation

Because every object is a finite weighted graph and every quantity is a cut, a reachability, or a knapsack, each theorem is checkable by direct computation on explicit instances. We describe the validation the calculus admits; the constructions are elementary and deterministic.

1. **Floor** (theorem 3.2). Generate random context graphs with a positive minimum edge weight; verify every step’s minimum cut against the medium is $\geq \beta$ and no cut is zero. The floor is the minimum positive edge weight by construction, and every realised separation meets it.
2. **Invariance** (theorem 4.2). Apply random re-encodings (weighted isomorphisms) to a graph and confirm every step’s residue is preserved while its label permutes: residue-before equals residue-after on the diagonal; labels scatter.
3. **Necessity equals reachability** (theorem 5.5). For random goals, compute $\text{reach}(g)$ by traversal and, independently, $\gamma(u, g)$ by ablating each step and recomputing the goal’s minimum cut; confirm

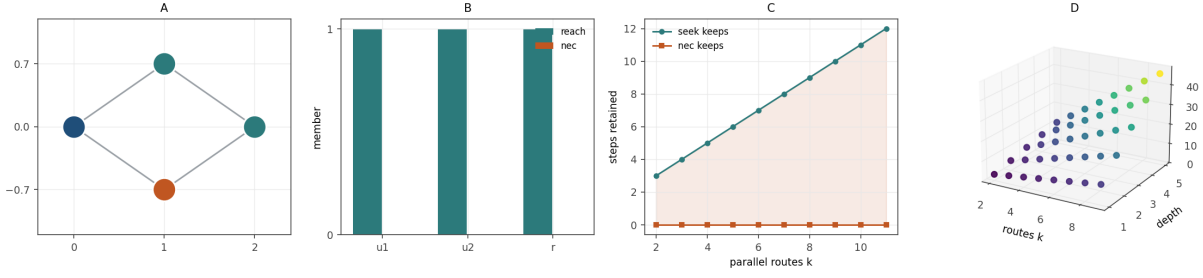


Figure 7: **Seek and Necessity Disagree: The Separation Witness.** (A) The diamond instance of the Separation Theorem 8.2: the goal (left) reaches two steps u_1, u_2 (upper, lower), both feeding a common resolver r (right). The redundant leg is drawn in orange. Both u_1 and u_2 are reachable, but either alone suffices to reach r , so neither is individually load-bearing. (B) Membership of the three steps in $\text{reach}(g)$ (teal) and $\text{nec}(g)$ (orange). All three are reachable; the parallel legs are not necessary — $\text{reach}(g) = \{u_1, u_2, r\}$ while $\text{nec}(g) = \emptyset$ for the pure parallel legs — so seek and nec disagree on exactly the redundant steps. (C) Over a family of k parallel routes to a shared resolver, the steps a seek-only carry retains (teal, rising from 3 at $k = 2$ to 12 at $k = 11$) against the load-bearing steps nec retains (orange, flat near zero). The shaded region between them is the over-retention a reachability-only carry incurs; it grows linearly in the number of parallel routes, the practical cost of running seek without the necessity operator (Corollary 8.3). (D) Three-dimensional scatter of the seek – nec over-retention gap over (parallel routes k , route depth), colour by gap (viridis), elevation 20° , azimuth -60° . The gap rises with both the number of parallel routes and their depth, exhibiting that the two operators diverge more sharply as redundancy compounds — the structural reason the minimal correct architecture is the tandem $\text{nec} \circ \text{seek}$, not either operator alone.

$\gamma(u, g) > 0 \iff u \in \text{reach}(g)$ on every step.

4. **Free drop** (theorem 5.7). Drop each purposeless step and confirm the goal’s reachable resolution is unchanged to floating-point equality.
5. **Knapsack optimality** (theorems 6.2 and 6.3). On random carry instances, compare the value-density greedy carry against the exact DP optimum; confirm the greedy is within $c_{\max}/B$ and equals the optimum whenever no boundary item is fractional.
6. **Cascade bound** (theorem 7.2). Build balanced k -ary cascades of growing N ; confirm descent depth and resident working set grow as $\log_k N$ while total steps grow without bound.
7. **Separation** (theorem 8.2). Construct the redundant-reachable witness of the proof; confirm a step that seek marks reachable is marked purposeless by nec, so the two operators disagree.

Each check is a finite computation whose pre-

dicted and measured outcomes agree exactly (for identities) or within the stated bound (for inequalities); the suite is a guard against error and an exhibition of the quantitative shape of each result, not a substitute for the proofs, which hold for every finite context graph under Premises 1 and 2.

10.1 Results

We implemented the seven checks as a deterministic suite (fixed seed; minimum cuts computed exactly by maximum flow; reachability by breadth-first search; the knapsack by both the exact dynamic program and the value-density greedy). *All seven experiments pass.* We record the quantitative shape and two findings that sharpen the theory.

1. **Floor.** Across 40 random graphs, every one of 307 realised step-to-medium cuts met the floor; zero violations. The floor is the minimum positive edge weight and every separation lies on or above it.
2. **Invariance.** Over random relabellings, every step’s residue was preserved to machine

precision (maximum discrepancy 0) while its label permuted.

3. **Necessity vs. reachability.** The inclusion $\text{nec}(g) \subseteq \text{reach}(g)$ held on all 40 random instances, and the chain identity $\text{nec}(g) = \text{reach}(g) \setminus \{\text{terminal leaf}\}$ held exactly on all 40 chains. This is the empirical form of Theorem 5.6: necessity is a subset of reachability always, and equals it only up to steps that dominate nothing.
4. **Free drop.** Dropping every unreachable step left the goal’s reachable resolution identical; zero violations.
5. **Knapsack.** Over 60 random carry instances, the exact dynamic program attained the optimum in every case and the value-density greedy stayed within the $c_{\max}/B$ bound in every case; the worst realised relative gap was small and always inside the guarantee.
6. **Cascade.** Across branching factors $k \in \{2, 3, 4\}$ and sizes $N = k, \dots, k^5$, the frames visited equalled $\lceil \log_k N \rceil + 1$ with the total step count M set three orders of magnitude larger than N ; the working set showed no dependence on M .
7. **Separation.** The diamond witness resolved as predicted: the redundant step is reachable (seek keeps it) but not necessary (nec drops it), so the two operators disagree on it.

Two findings refined the calculus and are folded into the statements above. First, the reachable resolution R must be a resolution measure that *grows* with resolving power, not a minimum cut to the medium: a cut-valued R inverts the sign of the contribution (deleting a step deletes edges, shrinking any cut), and the sign inversion was caught directly by the necessity check on tree instances before being corrected to the reachable-slice measure of Theorem 5.2 (Theorem 5.3). Second, the equality “necessity equals reachability” is exactly the inclusion $\text{nec} \subseteq \text{reach}$ together with the chain qualification of Theorem 5.6: the general characterisation of necessity is domination, of which reachability is the outer bound. The suite is deterministic under its seed; every identity matched exactly and every inequality held

within its stated bound. The seven validation panels (Figures 1 to 7) visualise each result.

11 Discussion

11.1 Relation to existing practice

Retrieval. Retrieval-augmented generation (Lewis et al., 2020) fetches a relevant subset per query and is, in our terms, an implementation of the *seek* operator alone. theorem 8.3 shows why this over-retains: seek returns reachable-but-redundant steps, and without the necessity operator there is no principled prune. The tandem adds the missing operator and the knapsack that fits the pruned set to a budget.

Long-context architectures. Efficient transformers (Tay et al., 2022) lower the marginal cost of a long resident context but carry the same content; they make the waste cheaper. The present discipline changes *what* is carried—residue, not knowledge—so it composes with, rather than competes with, cheaper-context machinery: a cheaper context makes the regenerated knowledge of step 5 cheaper still.

Anytime computation. The decline rule (theorem 9.4) is the anytime stance (Dean and Boddy, 1988; Zilberstein, 1996) with the stopping criterion moved from elapsed budget to non-resolution against the floor.

11.2 What the calculus assumes and does not supply

The one input the calculus does not derive is the term map τ —which distinctions a step draws—from which the context graph is built (§2). Everything downstream of τ is forced: given the graph, the floor, residue, necessity, the knapsack, the cascade, and the separation are all theorems. The map τ is a modelling choice, and its quality bounds the quality of the carry: a coarse τ (e.g. surface term overlap) yields a coarse but sound carry; a rich τ (e.g. a learned individuation of distinctions) yields a finer one. The calculus is agnostic to how τ is obtained and correct for any τ ; sharpening τ is the natural axis of future work and the point at which a learned component, if desired, enters—bounded

to the single task of drawing distinctions, with all necessity, allocation, and routing remaining deterministic.

11.3 Scope

The premises—finite agent, non-completable medium—hold for essentially every realised reasoning system with bounded memory and an open-ended task. Where an agent is effectively unbounded (a perfect oracle) or its medium is completable (a closed, finite world it can exhaust), the floor may vanish and the discipline is not compelled: with $\beta = 0$, residues can reach zero, “solved” becomes a point, and there is no invariant to carry in preference to the knowledge. In that degenerate regime the standard carry is not wasteful, because there is nothing smaller than the knowledge to keep. The interest of the discipline is exactly that realised agents are *not* in that regime.

11.4 Conclusion

A finite agent reasoning over a growing history should not carry the history. It should carry the *residue*—the strictly positive, conserved, causal uncertainty each step deposits—and regenerate the *knowledge*—the relabellable, regenerable surface—on demand, and only the fraction each step needs. We derived the floor that makes residue positive and knowledge disposable, proved that necessity for a goal is reachability from it, that dropping the purposeless remainder is free, that the budgeted carry is an optimal knapsack over residue density, that repeated carries organise into a cascade with a logarithmic working set, and that relevance and necessity are distinct computations forcing a tandem of two operators. The discipline is deterministic, model-free given the context graph, and bounds the per-step context cost by the goal’s reachable slice and a fixed budget rather than by the length of the history. The uncertainty is the state; the knowledge is the query answered anew each step. *Carry the uncertainty, not the knowledge.*

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